

Data fusion for heterogeneous survival effects: goals, role of the HDP error, and positioning

Working notes

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Scope

These notes accompany the proposed reframing of the project around two explicit goals:

- (G1) efficiency gain for heterogeneous treatment effect (HTE) estimation by combining a randomised trial (RCT, $S = 1$) with an observational study (OS, $S = 0$);
- (G2) identifiable extrapolation of causal survival contrasts beyond the RCT follow-up horizon t_{RCT} , using the longer follow-up of the OS.

Both goals are pursued under the AFT decomposition

$$Y_i = m_0(X_i, S_i) + \tau(X_i) A_i + (1 - S_i) A_i c(X_i) + \sigma \varepsilon_i, \quad Y_i = \log T_i, \quad (1)$$

with $m_0(X, S) = \mu(X) + (1 - S)g(X)$ in the four-forest version, and $\varepsilon \sim P$ with P assigned a hierarchical Dirichlet process (HDP) prior across data sources.

1 Pros of the reframing

Pro 1: a clinically named problem, not just a methodological one. “Better HTE precision” is a methodologist’s pitch. “Answering the long-term survival question that the trial cannot answer alone” is a clinical pitch that names a concrete gap routinely encountered in health technology assessment (HTA), oncology drug evaluation, and policy decision-making. The gap is well-established: pivotal RCTs typically report 2–5 years of follow-up, whereas registry data (NCDB, SEER, CPRD, claims databases) carry 10–20+ years (Latimer, 2013; Jackson, 2023; Guyot et al., 2017).

Pro 2: the AFT shift property yields a natural division of labour. On the log-time scale, $\tau(X)$ enters (1) as a constant covariate-specific shift. Once identified from short follow-up, the same shift applies for all t . This makes the two information channels conceptually orthogonal:

- the RCT identifies the *shift* $\tau(X)$ (and protects τ against confounding via the identification result; see Proposition 3 of the framework draft);

- the OS identifies the *shape* of the log-time distribution at long t (through the error CDF Φ) and the baseline m_0 .

The two sources therefore contribute complementary information, and not merely additional sample size.

Pro 3: the HDP is no longer a robustness garnish; it carries extrapolation. Survival extrapolation is exquisitely sensitive to the assumed error distribution. In a parametric AFT framework, lognormal, Weibull, log-logistic, Gompertz, and generalised gamma families fit short-term data equivalently well but yield wildly different tail behaviour, and consequently wildly different RMST values evaluated past the trial horizon. The NICE DSU TSD 14 workflow exists precisely because picking a tail family by AIC on RCT data alone is unsafe (Latimer, 2013; Bell Gorrod et al., 2019). The HDP prior on ε *is the mechanism* by which OS follow-up informs the tail: the shared atoms accumulate mass where both sources have data; OS-only mass shapes the tail; the hierarchical structure prevents source-specific drift from contaminating the shared body of the error distribution.

Pro 4: principled uncertainty quantification. Long-term extrapolations from parametric models often produce point estimates with spuriously narrow credible intervals, because the family is treated as known. A nonparametric error distribution propagates the genuine uncertainty about tail shape into the posterior of $\Delta_{\text{RMST}}(t^*, X)$ for $t^* > t_{\text{RCT}}$. As Jackson (2023) put it: long-term estimates should only be confident if there are strong long-term data. The HDP makes this property automatic.

Pro 5: the two goals share one estimation machinery. The same posterior over $(\mu, g, \tau, c, P, \sigma)$ yields both the HTE estimate and the extrapolated survival contrast. There is no two-stage hand-off, no plug-in step where uncertainty is silently dropped, and the confounding correction is in place throughout.

2 Challenges that need to be addressed

Challenge 1: constancy of the AFT shift over time. Under (1), $\tau(X)$ is a single number per covariate value; extrapolation assumes the shift identified from $[0, t_{\text{RCT}}]$ applies on (t_{RCT}, ∞) . This is a strong structural assumption. Treatments whose effect wanes (or grows) over time violate it, and the literature has explicit modelling devices for this case (waning treatment effects in Jackson (2023); time-varying hazard ratios more generally). Two paths forward:

- retain constant $\tau(X)$ and present this as a transparent assumption with sensitivity analyses;
- generalise to $\tau(X, t)$ via, e.g., a piecewise or spline-coefficient extension. This is feasible but breaks the clean “shift” interpretation.

Challenge 2: constancy of the confounding function over time. The decomposition (1) treats $c(X)$ analogously to $\tau(X)$: a covariate-specific log-time shift that does not vary with t . If selection bias has a different time signature than the treatment effect itself — e.g., frailty-driven early mortality differs across treatment arms in the OS but long-term survival does not — a single $c(X)$ is mis-specified. Worth flagging, with sensitivity to time-varying confounding.

Challenge 3: period and calendar-time effects in the OS. Long OS follow-up necessarily mixes historical cohorts treated under earlier standards of care. The patients contributing to the OS tail are *older calendar* patients, not just *older follow-up time*. Without adjustment, this can violate Assumption 4 (transportability) for reasons unrelated to unmeasured confounding. Three options: condition on calendar year as a covariate; restrict the OS window; or model an explicit period effect alongside g .

Challenge 4: informative censoring in the OS. Trial censoring is largely administrative and credibly non-informative; OS censoring includes loss-to-follow-up that may correlate with prognosis. The HDP error does not fix this — it assumes the residuals you observe at long times are exchangeable within source. A sensitivity analysis under a known censoring mechanism (or inverse-probability-of-censoring weighting) seems necessary.

Challenge 5: who anchors the shared atoms? With a fully pooled DP, the OS would dominate the error distribution (because it is typically much larger and runs longer). With fully separate DPs, the borrowing that gives extrapolation its power is lost. The *hierarchical* DP threads this needle, but the concentration parameters governing how aggressively the source-specific distributions deviate from the shared base must be either elicited or estimated. This is the extrapolation analogue of the commensurate-prior shrinkage on g controlled by k_g .

Challenge 6: identification of c relies on the RCT. From Proposition 3(iii) of the framework draft, the OS alone identifies $\tau + c$, not the two separately; the separation is anchored by the RCT. With a small RCT and a large OS, the prior on c does substantial work. Combined with extrapolation, this means: the RCT is doing *two* jobs simultaneously — anchoring the shift and disambiguating τ from c . A small RCT may not be able to do both well. Worth exploring via simulations that vary $n_{\text{RCT}}/n_{\text{OS}}$.

Challenge 7: covariate overlap at long times. The HTE $\tau(X = x)$ is informative only at covariate values x supported by both sources. For extrapolation, an additional implicit overlap is required: the OS must contain survivors at long t *with covariate profiles matching the RCT*. If the OS long-followup is dominated by, say, elderly patients who were not represented in the trial, the extrapolation for the trial-representative population is effectively pure prior.

Challenge 8: connecting target-population definitions across goals. Under (G1), the natural target is the OS population (representativeness gain). Under (G2), the target is whichever population the HTA decision concerns — often the trial population, projected forwards in time. The decomposition supports both, but the paper should be explicit about which target population is used in each numerical result, since the survival estimands in equations (23)–(26) of the framework draft contain c only when the target is the OS.

3 Possible extensions

(E1) **Time-varying treatment effect on the log-time scale.** Replace $\tau(X)$ by $\tau(X, t)$ via a spline or piecewise structure. This is the AFT-side analogue of waning treatment-effect models (Jackson, 2023).

- (E2) **Cure-fraction extension.** For oncology and immune-therapy settings, a mixture cure component is biologically motivated and is now standard in HTA extrapolation practice. Combining cure modelling with the four-forest decomposition is feasible and would broaden applicability.
- (E3) **Relative-survival / additive-hazards formulation.** Anchor the long-term hazard to known population mortality (life tables), so OS-tail information competes against a hard external constraint rather than against the prior alone. Particularly natural for older populations with substantial competing mortality.
- (E4) **Sensitivity analysis on unmeasured confounding.** Allow $c(X)$ to be informed by a prior derived from negative controls, or perform a tipping-point analysis over the magnitude of c consistent with observed RCT-OS discrepancy.
- (E5) **Multiple data sources.** Generalise from $S \in \{0, 1\}$ to a categorical source indicator, with the HDP base distribution shared across all sources and source-specific deviations regulated hierarchically. Several registries plus a trial; or a trial, an EHR cohort, and a claims database.
- (E6) **Aggregate external evidence.** Allow inclusion of summary survival data (e.g., from published KM curves reconstructed via Guyot et al. (2012)) alongside individual-level data, in the spirit of survextrap. Useful when long-term IPD are unavailable but published curves exist.
- (E7) **Posterior projection summaries for the HTE.** Map the high-dimensional posterior over $\tau(X)$ into an interpretable summary (a CART tree, a small set of subgroup rules) via the projection approach of Woody et al. (2020), and report how those subgroup summaries shift when the OS is included. This directly answers “how did borrowing change the heterogeneity story?”
- (E8) **Calendar-time covariate.** Adjust for cohort year inside the BART forests, mitigating Challenge 3.
- (E9) **Decision-analytic outputs.** Pipe the posterior of $\Delta_{\text{RMST}}(t^*, X)$ into life-years gained and (with utility weights) QALY contrasts. This is what HTA agencies actually want; the method already produces it.

4 Related work

The reframing places the project at the intersection of three literatures that have historically not communicated very much: causal data fusion, Bayesian survival extrapolation for HTA, and flexible Bayesian survival modelling. We sketch each below and indicate where the proposed method sits.

4.1 Survival extrapolation for HTA

The foundational reference is NICE DSU Technical Support Document 14 (Latimer, 2013), which codified the practice of fitting standard parametric AFT/PH families (exponential, Weibull, log-logistic, lognormal, Gompertz, generalised gamma) to trial data and selecting among them by fit and clinical plausibility. The TSD 14 process is widely followed but also widely criticised: fits in

the observed window are nearly indistinguishable, while extrapolations diverge dramatically (Bell Gorrod et al., 2019; Bagust and Beale, 2014).

External data have been used in HTA extrapolation in several distinct ways:

- **General-population mortality** as a lower bound on the hazard via additive-hazards (relative survival) models;
- **Registry conditional survival** from cancer registry databases used as external survival information for cancer (Guyot et al., 2017);
- **Expert elicitation** treated as prior data on long-term hazard (Guyot et al., 2017);
- **Longer-follow-up surrogate trials** such as the cilta-cel/LEGEND-2 example (Vyas et al., 2023), in which a phase I study with longer follow-up informs extrapolation of a pivotal phase II trial through informative priors on shape parameters.

The most directly comparable methodological piece is Jackson (2023)’s **survextrap**, which fits an M-spline hazard with a hierarchical prior on the coefficients, jointly to individual-level RCT data and external (possibly aggregate) data using Bayesian multi-parameter evidence synthesis. **survextrap** supports proportional and flexible non-proportional hazards, waning treatment effects, cure models, and additive-hazards formulations (Jackson, 2023). Empirical evaluations (Timmins et al., 2025) report that the model gives accurate long-term extrapolations when long-term external data on the patient population of interest are included, and quantifies structural uncertainty appropriately when they are not.

Where the proposed method differs from **survextrap.** **survextrap** does not address unmeasured confounding in the external data; the external sources it accommodates are general-population mortality, registry conditional survival, and elicited beliefs — not a confounded observational study of the *same* treatment. The proposed framework adds an explicit confounding-function correction $c(X)$, allowing the external data to be a contemporaneous OS subject to non-randomised treatment assignment, not merely a background hazard reference. The modelling style also differs: M-spline hazards versus BART forests on AFT components, and a flexible parametric error versus an HDP nonparametric error.

4.2 Causal data fusion under unmeasured confounding

The methodological core — a confounding function $c(X)$ correcting OS contrasts that do not hold unconfoundedness — is established in Yang et al. (2025), who derive semiparametric efficiency bounds for joint estimation of τ and c . Kallus et al. (2018) propose the “experimental grounding” framework, in which an RCT supplies a bias function correcting an observational HTE estimator. Yao et al. (2024) extend data fusion to partial OS treatment information. van der Laan et al. (2024) develop adaptive TMLE for RCT–RWD fusion that adapts borrowing to detected bias.

The closest match to the proposed work is Ye et al. (2025), who develop an integrative AFT framework for high-dimensional RCT and RWD with censoring and hidden confounding, using Stute-weighted penalised regression with separate regularisation for τ and the confounding component. The proposed method is essentially a Bayesian-nonparametric counterpart to Ye et al. (2025): BART ensembles in place of penalised parametric models, nonparametric error in place

of a Gaussian or assumed AFT family, and commensurate-prior structure on the baseline split between sources.

A practical caveat from Lin et al. (2024): bias-correction methods for marginal ATE estimation in this literature have not delivered RMSE improvements over RCT-only analysis in their simulations, and the methods’ theoretical advantage is for the *heterogeneous* effect function rather than its average. The proposed framing targets HTE plus long-term extrapolation, both of which are tasks for which RCT-only inference is genuinely deficient — the latter, simply because it cannot be done from the trial alone.

4.3 Bayesian flexible survival modelling

The BART-based modelling of μ , g , τ , c rests on the Bayesian causal forest (BCF) line of work (Hahn et al., 2020), which introduced the prognostic/treatment-effect decomposition with separate prior scales that the four-forest version generalises. The HDP error builds on the centred stick-breaking mixture of Yang et al. (2010) and the wider Bayesian-nonparametric survival literature; the hierarchical extension specifically lets the source label index the group hierarchy, with shared atoms across sources.

4.4 Positioning summary

	Confounding correction	Long-term extrapolation	Nonparametric error
Latimer (2013) (TSD 14 workflow)	—	yes	—
Guyot et al. (2017)	—	yes	—
Jackson (2023) (survextrap)	—	yes	flexible
Ye et al. (2025)	yes	limited [†]	—
Yang et al. (2025)	yes	—	—
Kallus et al. (2018), Yao et al. (2024)	yes	—	—
This work	yes	yes	yes (HDP)

Table 1: Schematic positioning. [†]: Ye et al. (2025) targets HTE estimation in the joint AFT framework; long-term extrapolation is not its stated goal but is in principle compatible with the AFT structure.

The niche is clear: the HTA-pragmatic literature handles extrapolation flexibly but does not correct for confounding in the external source; the causal-fusion literature corrects for confounding but does not target extrapolation; and few methods combine the two with a fully nonparametric error distribution. The proposed method sits deliberately in that gap.

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